Digit-Recurrence Division and Square Root

A polished, self-contained tutorial

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Abstract

Digit-recurrence algorithms compute a quotient or a square root one digit at a time while maintaining a scaled residual. Their hardware appeal comes from redundant signed-digit sets: because adjacent digits have overlapping validity regions, the next digit can be selected from truncated information rather than from an exact, full-width residual.

This tutorial develops the method from first principles and keeps the algebra explicit. We begin with the general notion of redundancy, derive the exact recurrences for division and square root, prove the residual containment conditions, and then show how those exact inequalities are converted into practical lookup-table selectors driven by truncated operands. Detailed Radix-2 and Radix-4 examples are worked out for both division and square root, including the determination of the precision parameters α and β, the treatment of positive and negative divisors, and the startup exceptions that appear in the square-root case. The last part explains how the overlap cells used by the "basic" selectors can be resolved deterministically to obtain the "optimized" selectors shown in the accompanying figures.

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1 Introduction

Division and square root are among the slowest elementary arithmetic operations in hardware. Unlike addition, subtraction, and many multiplication architectures, they are naturally iterative. Digit-recurrence algorithms attack the problem by generating the result from most significant digit to least significant digit. At each iteration they do two things:

1. choose the next quotient digit q_{j+1} or root digit s_{j+1}, and
2. update a scaled residual R_{j+1}.

The critical path is the digit selector. A naive selector would need the exact residual at full precision, which in turn would require a full-width carry-propagating addition every cycle. SRT algorithms avoid this by using a redundant digit set. Because adjacent digits overlap in their validity regions, the selector may safely use only a truncated estimate of the residual (and, for square root, a truncated estimate of the partial root as well). In practical hardware, the residual can therefore stay in carry-save form and only a short carry-propagate adder is needed in front of a small PLA or lookup table.

Two normalization choices are used throughout this tutorial:

$$\text{division: } N \in \left(0, \frac{1}{4}\right), \qquad D \in \left[\frac{1}{2}, 1\right],$$
$$\text{square root: } X \in \left[\frac{1}{4}, 1\right].$$

These are slightly tighter than the broadest possible SRT assumptions, but they make the startup logic much cleaner. The mathematics of the recurring loop is unchanged; only the bookkeeping for the first step becomes simpler.

2 Notation and design conventions

Table 1: Notation used throughout the tutorial.

Symbol	Meaning
B	Radix of the recurrence. The worked examples use $B = 2$ and $B = 4$.
a	Maximum digit magnitude, so the digit set is $\{-a, -a+1, \ldots, a\}$.
ρ	Redundancy factor, $\rho = a/(B-1)$.
Q_j	Quotient approximation after j selected quotient digits.
S_j	Square-root approximation after j selected root digits.
R_j	Scaled residual. The precise definition differs between division and square root.
q_{j+1}	Quotient digit chosen at iteration $j+1$.
s_{j+1}	Root digit chosen at iteration $j+1$.
α	Number of fractional bits retained from the truncated residual estimate.
β	Number of fractional bits retained from the truncated divisor or partial-root estimate.
τ_j	Integer encoding of the truncated residual estimate used by the selector.
δ	Integer encoding of the truncated divisor estimate in division.
σ_j	Integer encoding of the truncated partial-root estimate in square root.
$\mathrm{Int}(\cdot)$	"Remove the binary point and interpret the result as an integer." For example, $\mathrm{Int}\,(4 \cdot 0.10) = 2$.

The tutorial uses two complementary views of digit selection.

Exact view. The recurrence is analyzed over the real-valued residual R_j and the exact divisor D or partial root S_j. This yields ideal selection intervals.

Truncated view. Hardware does not see the exact residual. Instead it sees a truncated residual estimate t_j, converted to an integer τ_j, together with a truncated estimate of the divisor or partial root. The selector therefore acts on a finite grid:

$$(\delta, \tau_j) \quad \text{for division,} \qquad (\sigma_j, \tau_j) \quad \text{for square root.}$$

The Python scripts in the repository enumerate this finite grid and plot the cells that admit one or more legal digits. The formulas given later for the "basic" selectors are compact descriptions of those enumerated cells.

3 Redundant digit sets and overlap

For radix B, SRT-style digit recurrence selects every result digit from the symmetric set

$$q_{j+1}, s_{j+1} \in \{-a, -a+1, \ldots, a-1, a\}. \tag{3.1}$$

The redundancy factor is

$$\rho = \frac{a}{B-1}. \tag{3.2}$$

The key requirement is

$$\rho > \frac{1}{2}. \tag{3.3}$$

Why? Later we will derive intervals of the form

$$L_k \leq BR_j \leq U_k$$

that say when digit k is legal. Adjacent digits k and $k+1$ overlap exactly when $U_k \geq L_{k+1}$. That inequality reduces to $\rho \geq 1/2$, and the strict inequality $\rho > 1/2$ gives a nonzero overlap width. That nonzero overlap is what allows the selector to work from truncated information: if the exact residual lies near a boundary, either of the two adjacent digits can still keep the next residual inside the legal containment range.

Typical SRT designs choose

$$\frac{1}{2} < \rho \leq 1, \tag{3.4}$$

equivalently

$$\frac{B-1}{2} < a \leq B-1. \tag{3.5}$$

The upper end $\rho = 1$ is maximally redundant. Greater redundancy means wider overlap and easier selection, but it also means more multiples of the divisor (or of the partial root) must be formed. The designer therefore trades selector simplicity against datapath complexity.

4 Division

4.1 Normalized problem and quotient expansion

We start from

$$N = DQ + r, \tag{4.1}$$

where N is the dividend, D the divisor, Q the exact quotient, and r the final remainder.

With the tutorial normalization

$$N \in \left(0, \frac{1}{4}\right), \qquad D \in \left[\frac{1}{2}, 1\right],$$

the exact quotient satisfies

$$0 < \frac{N}{D} < \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}. \tag{4.2}$$

This is why we can safely start with $Q_0 = 0$ and choose the first digit from the nonnegative half of the digit set.

The quotient approximation after j digits is

$$Q_j = Q_0 + \sum_{i=0}^{j-1} q_{i+1} B^{-(i+1)}. \tag{4.3}$$

Equivalently,

$$Q_{j+1} = Q_j + q_{j+1} B^{-(j+1)}. \tag{4.4}$$

4.2 Tail bound and residual containment

Write the exact quotient as the infinite signed-digit expansion

$$\frac{N}{D} = Q_0 + \sum_{i=0}^{\infty} q_{i+1} B^{-(i+1)}. \tag{4.5}$$

Subtract eq. (4.3) from eq. (4.5). The approximation error after iteration j is

$$\varepsilon_j = \frac{N}{D} - Q_j = \sum_{i=j}^{\infty} q_{i+1} B^{-(i+1)}. \tag{4.6}$$

Because every digit satisfies $|q_{i+1}| \le a$,

$$
\begin{aligned}
|\varepsilon_j| &= \left| \sum_{i=j}^{\infty} q_{i+1} B^{-(i+1)} \right| \\
&\le \sum_{i=j}^{\infty} |q_{i+1}| \, B^{-(i+1)} \\
&\le \sum_{i=j}^{\infty} a B^{-(i+1)} \\
&= a B^{-(j+1)} \sum_{m=0}^{\infty} B^{-m} && (4.7) \\
&= a B^{-(j+1)} \cdot \frac{1}{1 - B^{-1}} \\
&= \frac{a}{B-1} B^{-j} \cdot B^{-1} \\
&= \rho B^{-j}. && (4.8)
\end{aligned}
$$

Replacing j with $j+1$ to match the selector indexing used later,

$$\left| \frac{N}{D} - Q_{j+1} \right| \le \rho B^{-(j+1)}. \tag{4.9}$$

Multiply eq. (4.9) by the positive quantity $B^{j+1}D$:

$$\left| B^{j+1}(N - DQ_{j+1}) \right| \le \rho D. \tag{4.10}$$

This motivates the scaled residual

$$R_{j+1} = B^{j+1}(N - DQ_{j+1}), \tag{4.11}$$

so the containment condition becomes

$$|R_{j+1}| \le \rho D. \tag{4.12}$$

At the end of the computation,

$$R_n = B^n(N - DQ_n) = B^n r. \tag{4.13}$$

4.3 Residual recurrence

Start from the definition of R_{j+1}:

$$
\begin{aligned}
R_{j+1} &= B^{j+1} \left(N - DQ_{j+1} \right) \\
&= B^{j+1} \left(N - D \left(Q_j + q_{j+1} B^{-(j+1)} \right) \right) && (4.14) \\
&= B^{j+1} \left(N - DQ_j \right) - q_{j+1} D \\
&= B \cdot B^j \left(N - DQ_j \right) - q_{j+1} D \\
&= B R_j - q_{j+1} D. && (4.15)
\end{aligned}
$$

So each cycle multiplies the old residual by the radix, subtracts the selected digit times the divisor, and must leave the new residual inside eq. (4.12).

4.4 Exact digit-selection intervals

4.4.1 Positive divisor

For $D > 0$, eq. (4.12) is

$$-\rho D \leq R_{j+1} \leq \rho D. \tag{4.16}$$

Insert eq. (4.15):

$$-\rho D \leq BR_j - q_{j+1}D \leq \rho D. \tag{4.17}$$

Add $q_{j+1}D$ throughout:

$$(q_{j+1} - \rho)\, D \leq BR_j \leq (q_{j+1} + \rho)\, D. \tag{4.18}$$

So, for a fixed positive divisor, digit k is legal exactly when

$$(k - \rho)D \leq BR_j \leq (k + \rho)D. \tag{4.19}$$

4.4.2 Negative divisor

For $D < 0$, $|D| = -D$, so eq. (4.12) becomes

$$\rho D \leq R_{j+1} \leq -\rho D. \tag{4.20}$$

Again insert eq. (4.15):

$$\rho D \leq BR_j - q_{j+1}D \leq -\rho D. \tag{4.21}$$

Add $q_{j+1}D$ to every term:

$$(q_{j+1} + \rho)D \leq BR_j \leq (q_{j+1} - \rho)D. \tag{4.22}$$

The endpoints appear reversed simply because D is negative.

4.5 Initialization

At $j = 0$, the approximation error bound is

$$\left| \frac{N}{D} - Q_1 \right| \leq \rho B^{-1}. \tag{4.23}$$

This is equivalent to

$$Q_1 - \rho B^{-1} \leq \frac{N}{D} \leq Q_1 + \rho B^{-1}. \tag{4.24}$$

With $Q_0 = 0$ and $Q_1 = q_1 B^{-1}$, the first digit must cover the entire interval $(0, \frac{1}{2})$ from eq. (4.2). It is enough that

$$Q_0 - \rho B^{-1} \leq 0, \tag{4.25}$$

$$\left(Q_0 + aB^{-1}\right) + \rho B^{-1} \geq \frac{1}{2}. \tag{4.26}$$

The first inequality is immediate because $Q_0 = 0$. In the second,

$$aB^{-1} + \rho B^{-1} = \frac{a + \rho}{B} = \frac{\rho(B - 1) + \rho}{B} = \rho.$$

Therefore eq. (4.26) reduces to $\rho \geq \frac{1}{2}$, which holds strictly in the redundant case.

The concrete startup intervals for the worked examples are:

Radix-2 ($B = 2, \rho = 1$)**.**

- If $q_1 = 0$, then $Q_1 = 0$ and

$$-\frac{1}{2} \le \frac{N}{D} \le \frac{1}{2} \quad \Longleftrightarrow \quad -\frac{1}{2}D \le N \le \frac{1}{2}D.$$

- If $q_1 = 1$, then $Q_1 = \frac{1}{2}$ and

$$0 \le \frac{N}{D} \le 1 \quad \Longleftrightarrow \quad 0 \le N \le D.$$

Radix-4 ($B = 4, \rho = \frac{2}{3}$)**.**

- If $q_1 = 0$, then $Q_1 = 0$ and

$$-\frac{1}{6} \le \frac{N}{D} \le \frac{1}{6} \quad \Longleftrightarrow \quad -\frac{1}{6}D \le N \le \frac{1}{6}D.$$

- If $q_1 = 1$, then $Q_1 = \frac{1}{4}$ and

$$\frac{1}{12} \le \frac{N}{D} \le \frac{5}{12} \quad \Longleftrightarrow \quad \frac{1}{12}D \le N \le \frac{5}{12}D.$$

- If $q_1 = 2$, then $Q_1 = \frac{1}{2}$ and

$$\frac{1}{3} \le \frac{N}{D} \le \frac{2}{3} \quad \Longleftrightarrow \quad \frac{1}{3}D \le N \le \frac{2}{3}D.$$

4.6 From exact intervals to a truncated selector

Hardware does not carry the exact value BR_j into the selector. Instead it keeps the residual in carry-save form and forms a short carry-propagated truncation. Let

$$BR_j = Y_j + Z_j,$$

with Y_j the carry-save sum word and Z_j the carry word. Keep only α fractional bits, add those retained bits with a small CPA, and call the result t_j. The one-sided truncation property is

$$0 \le BR_j - t_j < 2^{-(\alpha-1)}. \tag{4.27}$$

For division we also truncate the divisor to β fractional bits.

The selector itself does not operate on fixed-point fractions; it operates on integers. Hence we define

$$\tau_j = \mathrm{Int}\left(2^{\alpha} t_j\right), \tag{4.28}$$

$$\delta = \mathrm{Int}\left(2^{\beta} D_{\mathrm{trunc}}\right). \tag{4.29}$$

Once α and β are chosen, (δ, τ_j) lies on a finite grid. Each integer pair corresponds to a small rectangle in the exact (D, BR_j) plane. A digit is legal for that cell if some point in the cell satisfies the exact selector inequalities. The repository's Python scripts enumerate those cells and draw the legal regions. The table equations given later are just closed-form descriptions of those enumerated regions.

Before drawing the table, we still need a *continuity condition*: after truncation, adjacent digit regions must continue to overlap. That condition determines the minimum practical values of α and β.

Positive divisor continuity. For adjacent digits $k-1$ and k, define the exact upper and lower boundaries

$$U_{k-1}(D) = (k-1+\rho)D, \qquad L_k(D) = (k-\rho)D. \tag{4.30}$$

Residual truncation can reduce the usable upper boundary by at most $2^{-(\alpha-1)}$, so a sufficient overlap condition is

$$\min_D \big(U_{k-1}(D) - 2^{-(\alpha-1)}\big) \;\geq\; \max_D L_k(D). \tag{4.31}$$

Negative divisor continuity. For $D < 0$, the roles of the two boundary formulas reverse:

$$U_k(D) = (k-\rho)D, \qquad L_{k-1}(D) = (k-1+\rho)D. \tag{4.32}$$

The sufficient continuity condition is

$$\min_D \big(U_k(D) - 2^{-(\alpha-1)}\big) \;\geq\; \max_D L_{k-1}(D). \tag{4.33}$$

4.7 Worked example: Radix-2 division

Take $B = 2$ and the maximally redundant digit set $\{-1, 0, 1\}$. Then $a = 1$ and

$$\rho = \frac{1}{2-1} = 1. \tag{4.34}$$

4.7.1 Exact selection intervals

Substitute $B = 2$ and $\rho = 1$ into eq. (4.19) and eq. (4.22).

For $D > 0$,

$$q_{j+1} = \begin{cases} 1 & \text{if } 0 \leq 2R_j \leq 2D, \\ 0 & \text{if } -D \leq 2R_j \leq D, \\ -1 & \text{if } -2D \leq 2R_j \leq 0. \end{cases} \tag{4.35}$$

For $D < 0$,

$$q_{j+1} = \begin{cases} 1 & \text{if } 2D \leq 2R_j \leq 0, \\ 0 & \text{if } D \leq 2R_j \leq -D, \\ -1 & \text{if } 0 \leq 2R_j \leq -2D. \end{cases} \tag{4.36}$$

4.7.2 Choosing α and β

For the practical selector we use

$$\tau_j = \mathrm{Int}\left(2^{\alpha} t_j\right), \qquad \tau_j \leq 2^{\alpha+1} R_j < \tau_j + 2, \tag{4.37}$$

$$\delta = \mathrm{Int}\left(2^{\beta} D_{\mathrm{trunc}}\right), \qquad \delta \leq 2^{\beta} D < \delta + 1 \qquad (D > 0). \tag{4.38}$$

The smallest useful divisor truncation for a two-bin positive divisor is $\beta = 2$, so $\delta \in \{2, 3\}$.

Now apply the continuity condition.

Boundary between $q = 0$ and $q = 1$. Here $k = 1$. Since $\rho = 1$,

$$U_0(D) = (0 + 1)D = D, \qquad L_1(D) = (1 - 1)D = 0.$$

Insert these into eq. (4.31):

$$\begin{aligned} \min_D \left(D - 2^{-(\alpha-1)}\right) &\geq 0 \\ \frac{\delta}{2^\beta} - 2^{-(\alpha-1)} &\geq 0. \end{aligned} \tag{4.39}$$

Boundary between $q = -1$ and $q = 0$. Here $k = 0$. Again $\rho = 1$, so

$$U_{-1}(D) = (-1 + 1)D = 0, \qquad L_0(D) = (0 - 1)D = -D.$$

Insert these into eq. (4.31):

$$\begin{aligned} 0 - 2^{-(\alpha-1)} &\geq -D \\ D &\geq 2^{-(\alpha-1)} \\ \frac{\delta}{2^\beta} &\geq 2^{-(\alpha-1)}. \end{aligned} \tag{4.40}$$

The worst case is the smallest δ, namely $\delta_{\min} = 2^{\beta-1}$. With $\beta = 2$, eqs. (4.39) and (4.40) both reduce to

$$\frac{1}{2} \geq 2^{-(\alpha-1)},$$

hence the smallest admissible choice is

$$\alpha = 2, \qquad \beta = 2. \tag{4.41}$$

With that choice,

$$\tau_j = \mathrm{Int}\,(4\,t_j)\,, \qquad \tau_j \leq 8R_j < \tau_j + 2, \tag{4.42}$$

$$\delta = \mathrm{Int}\,(4D_{\mathrm{trunc}})\,, \qquad \delta \in \{2, 3\}\,. \tag{4.43}$$

4.7.3 Positive-divisor selector

Enumerating the finite grid (δ, τ_j) and marking every legal digit cell yields the basic positive-divisor selector

$$q_{j+1} = \begin{cases} 1 & \text{if } 0 \leq \tau_j < 2(\delta + 1), \\ 0 & \text{if } -\delta \leq \tau_j \leq \delta - 2, \\ -1 & \text{if } -2(\delta + 1) - 2 < \tau_j \leq -2. \end{cases} \tag{4.44}$$

Cells on the boundaries belong to two exact digit regions; the plot marks those cells as overlap cells.

4.7.4 Negative-divisor selector

For $D < 0$ we define

$$\tau_j = \mathrm{Int}\,(4\,t_j)\,, \qquad \tau_j \leq 8R_j < \tau_j + 2, \tag{4.45}$$

$$\delta = \mathrm{Int}\,(4D_{\mathrm{trunc}})\,, \qquad \delta \in \{-4, -3\}\,. \tag{4.46}$$

The continuity algebra is the sign-reflected version of the positive-divisor case, so the same choice $\alpha = 2, \beta = 2$ remains valid. Enumerating the (δ, τ_j) grid gives

$$q_{j+1} = \begin{cases} 1 & \text{if } 2\delta - 2 < \tau_j \le -2, \\ 0 & \text{if } \delta + 1 \le \tau_j \le -(\delta+1) - 2, \\ -1 & \text{if } 0 \le \tau_j \le -2\delta. \end{cases} \tag{4.47}$$

4.8 Worked example: Radix-4 division

Now take $B = 4$ with the minimally redundant digit set $\{-2, -1, 0, 1, 2\}$. Then $a = 2$ and

$$\rho = \frac{2}{4-1} = \frac{2}{3}. \tag{4.48}$$

4.8.1 Exact selection intervals

For $D > 0$, substitute $B = 4$ and $\rho = 2/3$ into eq. (4.19):

$$q_{j+1} = \begin{cases} 2 & \text{if } \frac{4}{3}D \le 4R_j \le \frac{8}{3}D, \\ 1 & \text{if } \frac{1}{3}D \le 4R_j \le \frac{5}{3}D, \\ 0 & \text{if } -\frac{2}{3}D \le 4R_j \le \frac{2}{3}D, \\ -1 & \text{if } -\frac{5}{3}D \le 4R_j \le -\frac{1}{3}D, \\ -2 & \text{if } -\frac{8}{3}D \le 4R_j \le -\frac{4}{3}D. \end{cases} \tag{4.49}$$

For $D < 0$,

$$q_{j+1} = \begin{cases} 2 & \text{if } \frac{8}{3}D \le 4R_j \le \frac{4}{3}D, \\ 1 & \text{if } \frac{5}{3}D \le 4R_j \le \frac{1}{3}D, \\ 0 & \text{if } \frac{2}{3}D \le 4R_j \le -\frac{2}{3}D, \\ -1 & \text{if } -\frac{1}{3}D \le 4R_j \le -\frac{5}{3}D, \\ -2 & \text{if } -\frac{4}{3}D \le 4R_j \le -\frac{8}{3}D. \end{cases} \tag{4.50}$$

4.8.2 Choosing α and β

Use

$$\tau_j = \mathrm{Int}\left(2^{\alpha} t_j\right), \qquad \tau_j \le 2^{\alpha+2} R_j < \tau_j + 2, \tag{4.51}$$

$$\delta = \mathrm{Int}\left(2^{\beta} D_{\mathrm{trunc}}\right), \qquad \delta \le 2^{\beta} D < \delta + 1 \qquad (D > 0). \tag{4.52}$$

The selector used in the scripts retains $\beta = 4$ fractional divisor bits, so $\delta \in \{8, 9, \ldots, 15\}$.

Apply eq. (4.31) to each adjacent digit boundary.

Boundary between $q = 1$ and $q = 2$. Here $k = 2$, so

$$U_1(D) = \left(1 + \frac{2}{3}\right) D = \frac{5}{3}D, \qquad L_2(D) = \left(2 - \frac{2}{3}\right) D = \frac{4}{3}D.$$

Therefore

$$\min_D \left(\frac{5}{3}D - 2^{-(\alpha-1)}\right) \geq \max_D \left(\frac{4}{3}D\right)$$
$$\frac{5}{3}\frac{\delta}{2^\beta} - 2^{-(\alpha-1)} \geq \frac{4}{3}\frac{\delta+1}{2^\beta}$$
$$\frac{1}{3}2^{-\beta}\delta - \frac{4}{3}2^{-\beta} - 2^{-(\alpha-1)} \geq 0. \tag{4.53}$$

Boundary between $q = 0$ and $q = 1$. Now $k = 1$, so

$$U_0(D) = \frac{2}{3}D, \qquad L_1(D) = \frac{1}{3}D.$$

Hence

$$\frac{2}{3}\frac{\delta}{2^\beta} - 2^{-(\alpha-1)} \geq \frac{1}{3}\frac{\delta+1}{2^\beta}$$
$$\frac{1}{3}2^{-\beta}\delta - \frac{1}{3}2^{-\beta} - 2^{-(\alpha-1)} \geq 0. \tag{4.54}$$

Boundary between $q = -1$ and $q = 0$. Now $k = 0$, so

$$U_{-1}(D) = -\frac{1}{3}D, \qquad L_0(D) = -\frac{2}{3}D.$$

The same algebra gives

$$\frac{1}{3}2^{-\beta}\delta - \frac{1}{3}2^{-\beta} - 2^{-(\alpha-1)} \geq 0. \tag{4.55}$$

Boundary between $q = -2$ and $q = -1$. Now $k = -1$, so

$$U_{-2}(D) = -\frac{4}{3}D, \qquad L_{-1}(D) = -\frac{5}{3}D.$$

Hence

$$\frac{1}{3}2^{-\beta}\delta - \frac{4}{3}2^{-\beta} - 2^{-(\alpha-1)} \geq 0. \tag{4.56}$$

The restrictive inequalities are eqs. (4.53) and (4.56). At the worst-case divisor bucket $\delta_{\min} = 8$ and $\beta = 4$,

$$\frac{1}{3}2^{-4} \cdot 8 - \frac{4}{3}2^{-4} = \frac{1}{6} - \frac{1}{12} = \frac{1}{12}.$$

So we need

$$\frac{1}{12} - 2^{-(\alpha-1)} \geq 0.$$

The smallest integer α satisfying this is $\alpha = 5$, because

$$2^{-(5-1)} = \frac{1}{16} \leq \frac{1}{12}, \qquad 2^{-(4-1)} = \frac{1}{8} > \frac{1}{12}.$$

Thus the design point used in the scripts is

$$\alpha = 5, \qquad \beta = 4. \tag{4.57}$$

With that choice,

$$\tau_j = \mathrm{Int}\left(32\, t_j\right), \qquad \tau_j \leq 128 R_j < \tau_j + 2, \tag{4.58}$$
$$\delta = \mathrm{Int}\left(16 D_{\mathrm{trunc}}\right), \qquad \delta \in \{8, \ldots, 15\}. \tag{4.59}$$

4.8.3 Positive-divisor selector

Enumerating the finite grid and fitting the resulting regions gives

$$q_{j+1} = \begin{cases} 2 & \text{if } \frac{8}{3}(\delta+1) \le \tau_j < \frac{16}{3}(\delta+1), \\ 1 & \text{if } \frac{2}{3}(\delta+1) \le \tau_j \le \frac{10}{3}\delta - 2, \\ 0 & \text{if } -\frac{4}{3}\delta \le \tau_j \le \frac{4}{3}\delta - 2, \\ -1 & \text{if } -\frac{10}{3}\delta \le \tau_j \le -\frac{2}{3}(\delta+1) - 2, \\ -2 & \text{if } -\frac{16}{3}(\delta+1) - 2 < \tau_j \le -\frac{8}{3}(\delta+1) - 2. \end{cases} \tag{4.60}$$

4.8.4 Negative-divisor selector

For $D < 0$, use

$$\tau_j = \mathrm{Int}\,(32\,t_j)\,, \qquad \tau_j \le 128R_j < \tau_j + 2, \tag{4.61}$$

$$\delta = \mathrm{Int}\,(16D_{\mathrm{trunc}})\,, \qquad \delta \in \{-16, \ldots, -9\}\,. \tag{4.62}$$

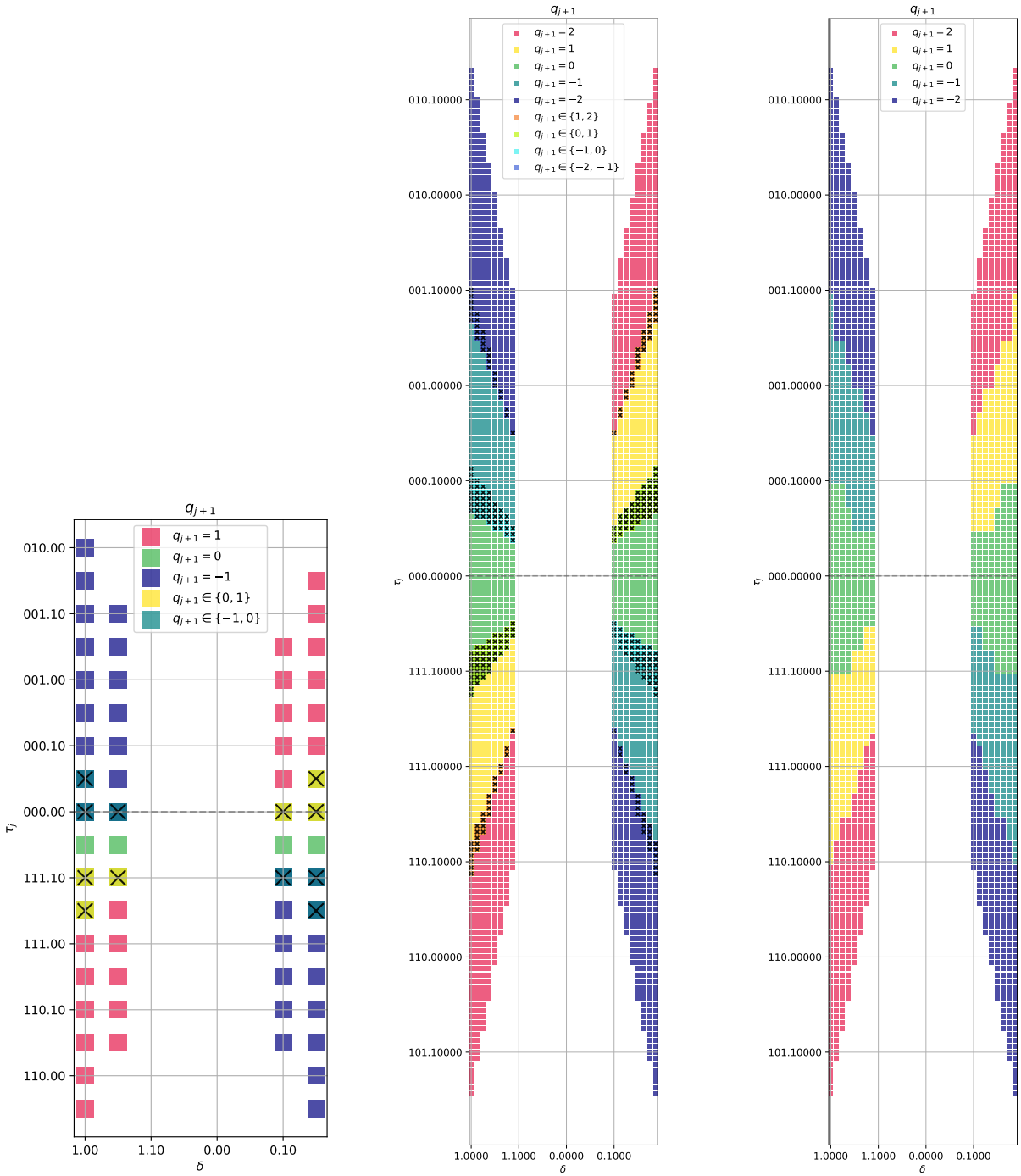
The sign-reflected cell enumeration yields

$$q_{j+1} = \begin{cases} 2 & \text{if } \frac{16}{3}\delta - 2 < \tau_j \le \frac{8}{3}\delta - 2, \\ 1 & \text{if } \frac{10}{3}(\delta+1) \le \tau_j \le \frac{2}{3}\delta - 2, \\ 0 & \text{if } \frac{4}{3}(\delta+1) \le \tau_j \le -\frac{4}{3}(\delta+1) - 2, \\ -1 & \text{if } -\frac{2}{3}\delta \le \tau_j \le -\frac{10}{3}(\delta+1) - 2, \\ -2 & \text{if } -\frac{8}{3}\delta \le \tau_j \le -\frac{16}{3}\delta. \end{cases} \tag{4.63}$$

4.9 Basic versus optimized division selectors

The *basic* selectors leave the overlap cells explicit: a cell may carry two legal digits, for example $\{0, 1\}$ or $\{-1, 0\}$. This is useful pedagogically because it makes the overlap visible. A hardware implementation usually goes one step further and assigns every overlap cell to a single digit. The optimized selectors used by the repository's Radix-4 scripts do exactly that.

The assignment is not arbitrary. Among all possible split points inside an overlap interval, the scripts choose split points aligned to convenient modulo classes (modulo 4, then 8, and then 16 as the overlap becomes wider). That alignment tends to simplify the resulting Boolean logic. The exact boundary-snap rule is given in appendix A. The important point for the main text is that the optimized selector never changes the exact legality of the recurrence; it only chooses one digit deterministically where the basic selector would have shown two.



(a) Radix-2 basic QDS (b) Radix-4 basic QDS (c) Radix-4 optimized QDS

Figure 1: Quotient-digit selector (QDS) regions in the truncated (δ, τ_j) plane. The basic selectors show overlap cells explicitly; the optimized selector resolves each overlap cell to a single digit using the rule described in appendix A.

5 Square root

5.1 Normalized problem and root expansion

The square-root problem is

$$X = S^2 + r, \tag{5.1}$$

where X is the radicand, $S = \sqrt{X}$ is the exact root, and r is the final remainder.

With the tutorial normalization $X \in [\frac{1}{4}, 1)$, the exact root lies in

$$\frac{1}{2} \le \sqrt{X} < 1. \tag{5.2}$$

This is why we can start from $S_0 = 1$: the initial approximation begins above the exact root and the selected digits bring it down toward the correct value.

The partial root after j digits is

$$S_j = S_0 + \sum_{i=0}^{j-1} s_{i+1} B^{-(i+1)}, \tag{5.3}$$

so

$$S_{j+1} = S_j + s_{j+1} B^{-(j+1)}. \tag{5.4}$$

5.2 Tail bound and residual containment

Write the exact root as the infinite signed-digit expansion

$$\sqrt{X} = S_0 + \sum_{i=0}^{\infty} s_{i+1} B^{-(i+1)}. \tag{5.5}$$

Subtract eq. (5.3) to obtain the error

$$e_j = \sqrt{X} - S_j = \sum_{i=j}^{\infty} s_{i+1} B^{-(i+1)}. \tag{5.6}$$

Exactly the same geometric-series bound used for division gives

$$|e_{j+1}| = \left|\sqrt{X} - S_{j+1}\right| \le \rho B^{-(j+1)}. \tag{5.7}$$

Now use the identity

$$X - S_{j+1}^2 = \left(\sqrt{X} - S_{j+1}\right)\left(\sqrt{X} + S_{j+1}\right) = e_{j+1}\left(2S_{j+1} + e_{j+1}\right). \tag{5.8}$$

Because e_{j+1} may be positive or negative, we bound both sides separately.

Upper bound. From $|e_{j+1}| \le \rho B^{-(j+1)}$ we have $e_{j+1} \le \rho B^{-(j+1)}$. Also $2S_{j+1} + e_{j+1} \le 2S_{j+1} + \rho B^{-(j+1)}$. Therefore

$$X - S_{j+1}^2 \le \rho B^{-(j+1)}\left(2S_{j+1} + \rho B^{-(j+1)}\right). \tag{5.9}$$

Expand the right-hand side:

$$X - S_{j+1}^2 \le 2\rho S_{j+1} B^{-(j+1)} + \rho^2 B^{-2(j+1)}. \tag{5.10}$$

Lower bound. Likewise $e_{j+1} \geq -\rho B^{-(j+1)}$, and

$$2S_{j+1} + e_{j+1} \geq 2S_{j+1} - \rho B^{-(j+1)}.$$

So

$$X - S_{j+1}^2 \geq -\rho B^{-(j+1)} \left(2S_{j+1} - \rho B^{-(j+1)}\right) \tag{5.11}$$

and therefore

$$X - S_{j+1}^2 \geq -2\rho S_{j+1} B^{-(j+1)} + \rho^2 B^{-2(j+1)}. \tag{5.12}$$

Collecting eqs. (5.10) and (5.12),

$$\rho^2 B^{-2(j+1)} - 2\rho S_{j+1} B^{-(j+1)} \leq X - S_{j+1}^2 \leq \rho^2 B^{-2(j+1)} + 2\rho S_{j+1} B^{-(j+1)}. \tag{5.13}$$

Define the scaled residual

$$R_{j+1} = B^{j+1} \left(X - S_{j+1}^2\right). \tag{5.14}$$

Multiply eq. (5.13) by B^{j+1} to obtain the containment condition

$$\rho^2 B^{-(j+1)} - 2\rho S_{j+1} \leq R_{j+1} \leq \rho^2 B^{-(j+1)} + 2\rho S_{j+1}. \tag{5.15}$$

At the end,

$$R_n = B^n \left(X - S_n^2\right) = B^n r. \tag{5.16}$$

5.3 Residual recurrence

Start from eq. (5.14) and use eq. (5.4):

$$\begin{aligned} R_{j+1} &= B^{j+1} \left(X - S_{j+1}^2\right) \\ &= B^{j+1} \left(X - \left(S_j + s_{j+1} B^{-(j+1)}\right)^2\right) \qquad (5.17) \\ &= B^{j+1} \left(X - S_j^2 - 2S_j s_{j+1} B^{-(j+1)} - s_{j+1}^2 B^{-2(j+1)}\right) \\ &= B^{j+1} \left(X - S_j^2\right) - 2S_j s_{j+1} - s_{j+1}^2 B^{-(j+1)} \\ &= BR_j - s_{j+1} \left(2S_j + s_{j+1} B^{-(j+1)}\right). \qquad (5.18) \end{aligned}$$

5.4 Exact digit-selection intervals

Insert eq. (5.18) into eq. (5.15). The left-hand side becomes

$$\begin{aligned} \rho^2 B^{-(j+1)} - 2\rho S_{j+1} &= \rho^2 B^{-(j+1)} - 2\rho \left(S_j + s_{j+1} B^{-(j+1)}\right) \\ &= -2\rho S_j + \left(\rho^2 - 2\rho s_{j+1}\right) B^{-(j+1)}. \end{aligned}$$

Move the subtraction term from eq. (5.18) to the left:

$$\begin{aligned} BR_j &\geq s_{j+1} \left(2S_j + s_{j+1} B^{-(j+1)}\right) + \rho^2 B^{-(j+1)} - 2\rho \left(S_j + s_{j+1} B^{-(j+1)}\right) \\ &= 2(s_{j+1} - \rho) S_j + \left(s_{j+1}^2 - 2\rho s_{j+1} + \rho^2\right) B^{-(j+1)} \\ &= 2(s_{j+1} - \rho) S_j + (s_{j+1} - \rho)^2 B^{-(j+1)}. \qquad (5.19) \end{aligned}$$

Exactly the same rearrangement on the upper side gives

$$BR_j \le 2(s_{j+1} + \rho)S_j + (s_{j+1} + \rho)^2 B^{-(j+1)}. \tag{5.20}$$

So the exact selection interval is

$$2(s_{j+1} - \rho)S_j + (s_{j+1} - \rho)^2 B^{-(j+1)} \le BR_j \le 2(s_{j+1} + \rho)S_j + (s_{j+1} + \rho)^2 B^{-(j+1)}. \tag{5.21}$$

This is the crucial difference from division: the interval boundaries depend on S_j, and they also contain explicit $B^{-(j+1)}$ terms. That extra iteration dependence is why square-root selectors need more careful startup treatment.

5.5 Initialization

At $j = 0$, the error bound is

$$\left|\sqrt{X} - S_1\right| \le \rho B^{-1}, \tag{5.22}$$

that is,

$$S_1 - \rho B^{-1} \le \sqrt{X} \le S_1 + \rho B^{-1}. \tag{5.23}$$

With $S_0 = 1$ and $S_1 = 1 + s_1 B^{-1}$, the first digit must cover the full interval $[\frac{1}{2}, 1)$ from eq. (5.2). The extreme requirements are

$$\left(S_0 - aB^{-1}\right) - \rho B^{-1} \le \frac{1}{2}, \tag{5.24}$$

$$S_0 + \rho B^{-1} \ge 1. \tag{5.25}$$

The second inequality is immediate from $S_0 = 1$. In the first,

$$aB^{-1} + \rho B^{-1} = \rho,$$

so eq. (5.24) reduces to $1 - \rho \le \frac{1}{2}$, again satisfied for $\rho \ge \frac{1}{2}$.

The explicit startup intervals are:

Radix-2 $(B = 2, \rho = 1)$**.**

- If $s_1 = 0$, then $S_1 = 1$ and
$$\frac{1}{2} \le \sqrt{X} \le \frac{3}{2} \quad \Longleftrightarrow \quad \frac{1}{4} \le X \le \frac{9}{4}.$$
- If $s_1 = -1$, then $S_1 = \frac{1}{2}$ and
$$0 \le \sqrt{X} \le 1 \quad \Longleftrightarrow \quad 0 \le X \le 1.$$

Radix-4 $(B = 4, \rho = \frac{2}{3})$**.**

- If $s_1 = 0$, then $S_1 = 1$ and
$$\frac{5}{6} \le \sqrt{X} \le \frac{7}{6} \quad \Longleftrightarrow \quad \frac{25}{36} \le X \le \frac{49}{36}.$$
- If $s_1 = -1$, then $S_1 = \frac{3}{4}$ and
$$\frac{7}{12} \le \sqrt{X} \le \frac{11}{12} \quad \Longleftrightarrow \quad \frac{49}{144} \le X \le \frac{121}{144}.$$
- If $s_1 = -2$, then $S_1 = \frac{1}{2}$ and
$$\frac{1}{3} \le \sqrt{X} \le \frac{2}{3} \quad \Longleftrightarrow \quad \frac{1}{9} \le X \le \frac{4}{9}.$$

5.6 From exact intervals to a truncated selector

Square root uses the same residual truncation idea as division, but the selector also needs a truncated partial-root estimate. Let t_j again denote the carry-propagated truncation of BR_j with α fractional bits, so

$$0 \le BR_j - t_j < 2^{-(\alpha-1)}. \tag{5.26}$$

Define

$$\tau_j = \mathrm{Int}\left(2^{\alpha} t_j\right), \tag{5.27}$$

$$\sigma_j = \mathrm{Int}\left(2^{\beta} S_{j,\mathrm{trunc}}\right). \tag{5.28}$$

As in division, the selector plane is finite: (σ_j, τ_j) ranges over finitely many cells.

The continuity condition is the square-root analogue of eq. (4.31), but now the exact boundaries depend on S_j and j. For adjacent digits $k-1$ and k, define

$$U_{k-1}(S_j, j) = 2(k-1+\rho)S_j + (k-1+\rho)^2 B^{-(j+1)}, \tag{5.29}$$

$$L_k(S_j, j) = 2(k-\rho)S_j + (k-\rho)^2 B^{-(j+1)}. \tag{5.30}$$

A sufficient continuity condition is

$$\min_{S_j}\left(U_{k-1}(S_j, j) - 2^{-(\alpha-1)}\right) \ \ge\ \max_{S_j} L_k(S_j, j). \tag{5.31}$$

Later we will apply this formula with specific Radix-2 and Radix-4 coefficients.

5.7 Worked example: Radix-2 square root

Take $B = 2$, $a = 1$, $\rho = 1$, and the digit set $\{-1, 0, 1\}$.

5.7.1 Exact selection intervals

Insert $B = 2$ and $\rho = 1$ into eq. (5.21):

$$s_{j+1} = \begin{cases} 1 & \text{if } 0 \le 2R_j \le 4S_j + 4\cdot 2^{-(j+1)}, \\ 0 & \text{if } -2S_j + 2^{-(j+1)} \le 2R_j \le 2S_j + 2^{-(j+1)}, \\ -1 & \text{if } -4S_j + 4\cdot 2^{-(j+1)} \le 2R_j \le 0. \end{cases} \tag{5.32}$$

5.7.2 Generic selector and precision choice

The selector uses

$$\tau_j = \mathrm{Int}\left(2^{\alpha} t_j\right), \qquad \tau_j \le 2^{\alpha+1} R_j < \tau_j + 2, \tag{5.33}$$

$$\sigma_j = \mathrm{Int}\left(2^{\beta} S_{j,\mathrm{trunc}}\right). \tag{5.34}$$

For the generic path (that is, excluding the startup exceptions discussed next), the scripts use $\beta = 2$ and a single partial-root bucket $\sigma_j = 3$.

Apply eq. (5.31).

Boundary between $s = 0$ and $s = 1$. Here $k = 1$. Since $\rho = 1$,

$$U_0(S_j, j) = 2S_j + 2^{-(j+1)}, \qquad L_1(S_j, j) = 0.$$

The positive constant $2^{-(j+1)}$ only helps, so the simpler sufficient inequality is

$$2 \cdot 2^{-\beta}\sigma_j - 2^{-(\alpha-1)} \geq 0. \tag{5.35}$$

Boundary between $s = -1$ and $s = 0$. Here $k = 0$, so

$$U_{-1}(S_j, j) = 0, \qquad L_0(S_j, j) = -2S_j + 2^{-(j+1)}.$$

Using the largest admissible S_j for a given σ_j gives the sufficient inequality

$$2 \cdot 2^{-\beta}\sigma_j - 2^{-(\alpha-1)} - 2^{-(j+1)} \geq 0. \tag{5.36}$$

With the generic bucket $\sigma_j = 3$, the smallest choice satisfying both inequalities is

$$\alpha = 2, \qquad \beta = 2. \tag{5.37}$$

Therefore

$$\tau_j = \mathrm{Int}\,(4\,t_j)\,, \qquad \tau_j \leq 8R_j < \tau_j + 2, \tag{5.38}$$

$$\sigma_j = 3. \tag{5.39}$$

The corresponding basic selector is

$$s_{j+1} = \begin{cases} 1 & \text{if } 0 \leq \tau_j < 4(\sigma_j + 1), \\ 0 & \text{if } -2(\sigma_j - 1) \leq \tau_j \leq 2\sigma_j - 2, \\ -1 & \text{if } -4(\sigma_j + 1) - 2 < \tau_j \leq -2. \end{cases} \tag{5.40}$$

5.7.3 Startup path near $X = \frac{1}{4}$

When X is close to $\frac{1}{4}$, initialization chooses $s_1 = -1$ and the following digits stay zero for a while:

$$s_1 = -1, \qquad s_2 = s_3 = \cdots = s_j = 0.$$

Since $S_1 = 1 - \frac{1}{2} = \frac{1}{2}$, the partial root remains

$$S_j = \frac{1}{2} \qquad (j \geq 1). \tag{5.41}$$

Hence the truncated partial-root bucket is the minimum one:

$$\sigma_j = \mathrm{Int}\,(4 \cdot 0.10) = 2. \tag{5.42}$$

The initial residual is

$$R_1 = 2\left(X - \frac{1}{4}\right) \geq 0.$$

Whenever the next digit is zero, the recurrence reduces to $R_k = 2R_{k-1}$, so the residual stays nonnegative throughout this zero-digit prefix. Since

$$0 \leq 2R_j - t_j < 2^{-(\alpha-1)},$$

we have $t_j > -2^{-(\alpha-1)}$, and with $\alpha = 2$,

$$\tau_j \geq -2. \tag{5.43}$$

Therefore the negative-digit region is unreachable and the selector collapses to

$$s_{j+1} = \begin{cases} 1 & \text{if } 0 \leq \tau_j < 4(\sigma_j + 1), \\ 0 & \text{if } -2 \leq \tau_j \leq 2\sigma_j - 2, \end{cases} \qquad \sigma_j = 2. \tag{5.44}$$

5.7.4 Startup path near $X = 1$

When X is close to 1, the startup path is

$$s_1 = s_2 = \cdots = s_j = 0.$$

Then

$$S_j = 1 \tag{5.45}$$

throughout the zero-digit prefix. Because $X < 1$,

$$R_1 = 2(X - 1) < 0,$$

and the zero-digit recurrence $R_k = 2R_{k-1}$ preserves that negativity. Hence $2R_j < 0$, and because truncation is one-sided, $t_j < 0$ as well. Therefore

$$\tau_j \leq -1. \tag{5.46}$$

The truncated partial-root bucket is

$$\sigma_j = \mathrm{Int}\,(4 \cdot 1.00) = 4. \tag{5.47}$$

The positive-digit region is unreachable, so the selector reduces to

$$s_{j+1} = \begin{cases} 0 & \text{if } -2(\sigma_j - 1) \leq \tau_j \leq -1, \\ -1 & \text{if } -4\sigma_j - 2 < \tau_j \leq -2. \end{cases} \tag{5.48}$$

5.8 Worked example: Radix-4 square root

Take $B = 4$, $a = 2$, $\rho = 2/3$, and digit set $\{-2, -1, 0, 1, 2\}$.

5.8.1 Exact selection intervals

Substitute $B = 4$ and $\rho = 2/3$ into eq. (5.21):

$$s_{j+1} = \begin{cases} 2 & \text{if } \frac{8}{3}S_j + \frac{16}{9}4^{-(j+1)} \leq 4R_j \leq \frac{16}{3}S_j + \frac{64}{9}4^{-(j+1)}, \\ 1 & \text{if } \frac{2}{3}S_j + \frac{1}{9}4^{-(j+1)} \leq 4R_j \leq \frac{10}{3}S_j + \frac{25}{9}4^{-(j+1)}, \\ 0 & \text{if } -\frac{4}{3}S_j + \frac{4}{9}4^{-(j+1)} \leq 4R_j \leq \frac{4}{3}S_j + \frac{4}{9}4^{-(j+1)}, \\ -1 & \text{if } -\frac{10}{3}S_j + \frac{25}{9}4^{-(j+1)} \leq 4R_j \leq -\frac{2}{3}S_j + \frac{1}{9}4^{-(j+1)}, \\ -2 & \text{if } -\frac{16}{3}S_j + \frac{64}{9}4^{-(j+1)} \leq 4R_j \leq -\frac{8}{3}S_j + \frac{16}{9}4^{-(j+1)}. \end{cases} \tag{5.49}$$

5.8.2 Generic selector for $j \geq 2$

Use

$$\tau_j = \mathrm{Int}\,(2^{\alpha} t_j)\,, \qquad \tau_j \leq 2^{\alpha+2} R_j < \tau_j + 2, \tag{5.50}$$

$$\sigma_j = \mathrm{Int}\left(2^{\beta} S_{j,\mathrm{trunc}}\right). \tag{5.51}$$

The scripts use $\beta = 4$, so the generic buckets are $\sigma_j \in \{9, \ldots, 15\}$. Apply eq. (5.31).

Boundary between $s = 1$ and $s = 2$. Here $k = 2$, so

$$U_1(S_j, j) = \frac{10}{3} S_j + \frac{25}{9} 4^{-(j+1)}, \qquad L_2(S_j, j) = \frac{8}{3} S_j + \frac{16}{9} 4^{-(j+1)}.$$

Using the smallest S_j on the upper side and the largest S_j on the lower side, the constant terms combine to the sufficient inequality

$$\frac{2}{3} 2^{-\beta} \sigma_j - \frac{8}{3} 2^{-\beta} - 2^{-(\alpha-1)} \geq 0. \tag{5.52}$$

Boundary between $s = 0$ and $s = 1$. This gives

$$\frac{2}{3} 2^{-\beta} \sigma_j - \frac{2}{3} 2^{-\beta} - 2^{-(\alpha-1)} \geq 0. \tag{5.53}$$

Boundary between $s = -1$ and $s = 0$. Now the explicit $4^{-(j+1)}$ term matters:

$$\frac{2}{3} 2^{-\beta} \sigma_j - \frac{2}{3} 2^{-\beta} - 2^{-(\alpha-1)} - \frac{4}{9} 4^{-(j+1)} \geq 0. \tag{5.54}$$

Boundary between $s = -2$ and $s = -1$. This is the tightest one:

$$\frac{2}{3} 2^{-\beta} \sigma_j - \frac{8}{3} 2^{-\beta} - 2^{-(\alpha-1)} - \frac{25}{9} 4^{-(j+1)} \geq 0. \tag{5.55}$$

Choose $\beta = 4$ and try $\alpha = 4$. At the worst generic bucket $\sigma_j = 9$,

$$\begin{aligned} \frac{2}{3} 2^{-4} \sigma_j - \frac{8}{3} 2^{-4} - 2^{-(4-1)} &= \frac{2}{3} \cdot \frac{9}{16} - \frac{8}{3} \cdot \frac{1}{16} - \frac{1}{8} \\ &= \frac{3}{8} - \frac{1}{6} - \frac{1}{8} \\ &= \frac{1}{12}. \end{aligned}$$

So eq. (5.55) becomes

$$\frac{1}{12} - \frac{25}{9} 4^{-(j+1)} \geq 0. \tag{5.56}$$

For $j = 1$, the left-hand side is

$$\frac{1}{12} - \frac{25}{9} \cdot \frac{1}{16} = \frac{4}{48} - \frac{25}{144} = -\frac{13}{144} < 0.$$

For $j = 2$, it is

$$\frac{1}{12} - \frac{25}{9} \cdot \frac{1}{64} = \frac{16}{192} - \frac{25}{576} = \frac{23}{576} > 0.$$

Therefore the generic $(\alpha, \beta) = (4, 4)$ selector is safe from $j \geq 2$ onward.

With that choice,

$$\tau_j = \mathrm{Int}\,(16\, t_j)\,, \qquad \tau_j \leq 64 R_j < \tau_j + 2, \tag{5.57}$$

$$\sigma_j \in \{9, \ldots, 15\}\,. \tag{5.58}$$

The resulting basic selector for $j \geq 2$ is

$$s_{j+1} = \begin{cases} 2 & \text{if } \frac{8}{3}(\sigma_j + 1) \leq \tau_j < \frac{16}{3}(\sigma_j + 1), \\ 1 & \text{if } \frac{2}{3}(\sigma_j + 1) \leq \tau_j \leq \frac{10}{3}\sigma_j - 2, \\ 0 & \text{if } -\frac{4}{3}\left(\sigma_j - \frac{1}{12}\right) \leq \tau_j \leq \frac{4}{3}\sigma_j - 2, \\ -1 & \text{if } -\frac{10}{3}\left(\sigma_j - \frac{5}{24}\right) \leq \tau_j \leq -\frac{2}{3}(\sigma_j + 1) - 2, \\ -2 & \text{if } -\frac{16}{3}(\sigma_j + 1) - 2 < \tau_j \leq -\frac{8}{3}(\sigma_j + 1) - 2. \end{cases} \tag{5.59}$$

5.8.3 Startup path near $X = \frac{1}{4}$

When X is close to $\frac{1}{4}$, initialization chooses $s_1 = -2$ and then stays at zero for a while:

$$s_1 = -2, \qquad s_2 = s_3 = \cdots = s_j = 0.$$

Since $S_1 = 1 - \frac{2}{4} = \frac{1}{2}$, the partial root remains

$$S_j = \frac{1}{2} \qquad (j \geq 1), \tag{5.60}$$

so

$$\sigma_j = \mathrm{Int}\,(16 \cdot 0.1000) = 8. \tag{5.61}$$

The initial residual is

$$R_1 = 4\left(X - \frac{1}{4}\right) \geq 0,$$

and zero digits preserve that nonnegativity. Since

$$0 \leq 4R_j - t_j < 2^{-(\alpha-1)},$$

we have $t_j > -2^{-(\alpha-1)}$. With $\alpha = 4$, that implies

$$\tau_j \geq -2. \tag{5.62}$$

So the negative-side overlap cells are unreachable, and the selector reduces to

$$s_{j+1} = \begin{cases} 2 & \text{if } \frac{8}{3}(\sigma_j + 1) \leq \tau_j < \frac{16}{3}(\sigma_j + 1), \\ 1 & \text{if } \frac{2}{3}(\sigma_j + 1) \leq \tau_j \leq \frac{10}{3}\sigma_j - 2, \\ 0 & \text{if } -2 \leq \tau_j \leq \frac{4}{3}\sigma_j - 2, \end{cases} \qquad \sigma_j = 8. \tag{5.63}$$

5.8.4 Startup path near $X = 1$

When X is close to 1, the startup path is

$$s_1 = s_2 = \cdots = s_j = 0,$$

so

$$S_j = 1. \tag{5.64}$$

Again $R_1 = 4(X - 1) < 0$, and zero digits preserve the sign. Therefore $4R_j < 0$, and the one-sided truncation implies

$$\tau_j \leq -1. \tag{5.65}$$

The partial-root bucket is now

$$\sigma_j = \mathrm{Int}\,(16 \cdot 1.0000) = 16. \tag{5.66}$$

Re-evaluate the restrictive continuity inequality eq. (5.55) at $\sigma_j = 16$, $\alpha = 4$, and $\beta = 4$:

$$\begin{aligned} \frac{2}{3}2^{-4} \cdot 16 - \frac{8}{3}2^{-4} - 2^{-(4-1)} - \frac{25}{9}4^{-(j+1)} &= \frac{2}{3} - \frac{1}{6} - \frac{1}{8} - \frac{25}{9}4^{-(j+1)} \\ &= \frac{3}{8} - \frac{25}{9}4^{-(j+1)}. \end{aligned} \tag{5.67}$$

For $j = 1$,

$$\frac{3}{8} - \frac{25}{9} \cdot \frac{1}{16} = \frac{54}{144} - \frac{25}{144} = \frac{29}{144} > 0.$$

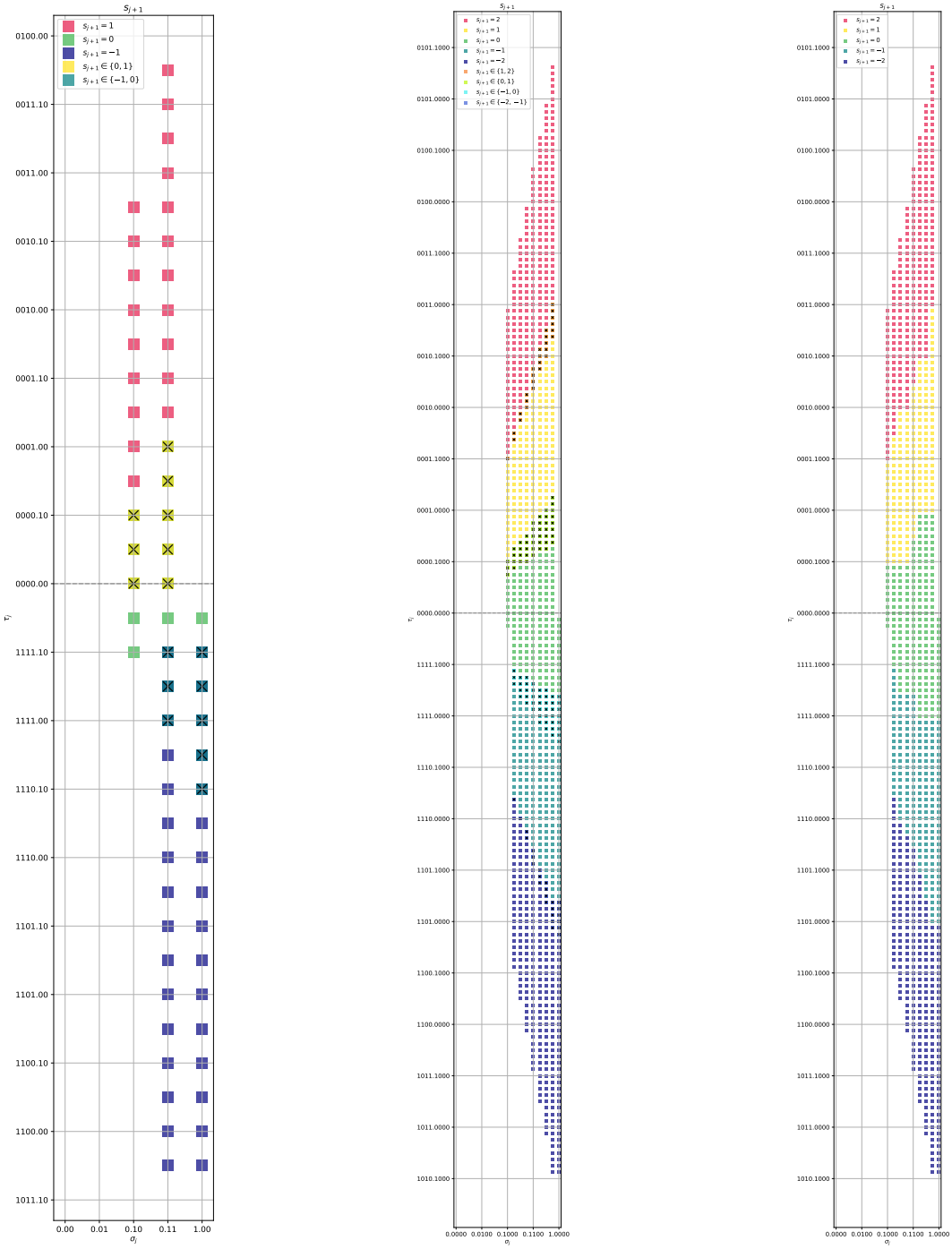
So this path is already safe from the first standard post-initialization iteration onward.

Because $\tau_j \leq -1$, the positive-side selector cells are unreachable. The reduced selector is

$$s_{j+1} = \begin{cases} 0 & \text{if } -\frac{4}{3}\left(\sigma_j - \frac{1}{3}\right) \leq \tau_j \leq -1, \\ -1 & \text{if } -\frac{10}{3}\left(\sigma_j - \frac{5}{6}\right) \leq \tau_j \leq -\frac{2}{3}\sigma_j - 2, \\ -2 & \text{if } -\frac{16}{3}\sigma_j - 2 < \tau_j \leq -\frac{8}{3}\sigma_j - 2. \end{cases} \tag{5.68}$$

5.9 Basic versus optimized square-root selectors

Exactly as in division, the basic Radix-4 square-root selector leaves overlap cells explicit, while the optimized selector resolves those overlap cells using the modulo-aligned boundary rule from appendix A. The optimized selector does not change the recurrence or its legal containment region; it only turns a multivalued overlap cell into a single deterministic choice.



(a) Radix-2 basic RDS (b) Radix-4 basic RDS (c) Radix-4 optimized RDS

Figure 2: Root-digit selector (RDS) regions in the truncated (σ_j, τ_j) plane. The Radix-4 basic selector makes the overlap cells explicit; the optimized selector resolves them deterministically.

6 Practical summary

The complete design workflow is the same for division and square root.

1. Choose a radix B and a redundant digit set $\{-a, \ldots, a\}$, hence $\rho = a/(B-1)$.
2. Derive the exact recurrence for the scaled residual R_j.
3. Derive the exact selection intervals that guarantee the next residual remains contained.
4. Introduce truncated selector variables τ_j and δ or σ_j.
5. Use a continuity condition to choose α and β large enough that adjacent digit regions still overlap after truncation.
6. Enumerate the finite selector grid and record which digits are legal in each cell.
7. Either:
   - keep the overlap cells explicit (basic selector), or
   - assign each overlap cell to one digit using a deterministic boundary rule (optimized selector).

Division is the easier of the two problems because the selection boundaries depend only on the divisor D. Square root is harder because the boundaries depend on the evolving partial root S_j and on iteration-dependent $B^{-(j+1)}$ terms. Those extra terms are precisely why the Radix-4 square-root example needs separate startup-path analysis.

7 Conclusion

Digit-recurrence arithmetic is built on a simple but powerful idea: use a redundant digit set so that adjacent digits overlap, and then exploit that overlap to perform fast selection from truncated state. The full-width exact residual is never required on the selector critical path.

In this tutorial we:

- derived the exact SRT-style recurrences for division and square root,
- proved the residual containment conditions from the infinite signed-digit expansions,
- converted the exact inequalities into practical truncated selectors,
- worked through Radix-2 and Radix-4 division,
- worked through Radix-2 and Radix-4 square root, including the startup special cases, and
- explained how the optimized selectors arise from deterministic overlap resolution.

The central design trade-off is clear. A higher radix can produce more result bits per iteration, but it typically requires more selector precision and more complicated multiples. More redundancy widens the legal overlap and simplifies selection, but it also enlarges the datapath work associated with each selected digit. SRT design is therefore not only about algebra; it is about choosing a comfortable operating point in this three-way trade-off among radix, redundancy, and selector complexity.

A Deterministic overlap resolution used by the optimized Radix-4 scripts

The basic selectors leave overlap cells visible. The optimized Radix-4 scripts replace each overlap interval by a single split point chosen to land on logic-friendly modulo boundaries. This appendix summarizes the exact rule implemented in the repository's `transform` helper.

Suppose two adjacent digits have the interval endpoints

$$\text{lower digit upper bound } = x, \qquad \text{higher digit lower bound } = y,$$

with $x \geq y$. The overlap width is measured by

$$\text{diff} = x - y.$$

The scripts replace the original pair (x, y) by a snapped pair (x', y') with $x' < y'$, thereby resolving the overlap to a single boundary.

Case 1: zero-width overlap ($\text{diff} = 0$**).** If the shared boundary is odd, choose the pair $(x, x+1)$. If it is even, choose $(x-1, x)$. This pushes the final split to an odd/even boundary that is easier to encode.

Case 2: small overlap ($1 \leq \text{diff} \leq 2$**).** Use modulus $M = 4$. Let

$$\text{base} = \left\lfloor \frac{y-1}{4} \right\rfloor \cdot 4.$$

Then the scripts choose either

$$(x', y') = (\text{base} + 1,\ \text{base} + 2)$$

or

$$(x', y') = (\text{base} + 3,\ \text{base} + 4)\,,$$

depending on whether the original upper endpoint x lies below or above the threshold $\text{base} + 3$.

Case 3: medium overlap ($3 \leq \text{diff} \leq 6$**).** Use modulus $M = 8$. With

$$\text{base} = \left\lfloor \frac{y-1}{8} \right\rfloor \cdot 8,$$

the snapped pair is either

$$(x', y') = (\text{base} + 3,\ \text{base} + 4)$$

or

$$(x', y') = (\text{base} + 7,\ \text{base} + 8)\,,$$

again depending on whether x is below or above the threshold $\text{base} + 7$.

Case 4: large overlap ($7 \leq \text{diff} \leq 14$). The same rule is applied with modulus $M = 16$, which snaps the split to a $(16a + 7, 16a + 8)$-type boundary.

In code form, the rule is:

```
diff = x - y
if diff == 0:  snap to (x,x+1) or (x-1,x) depending on parity
elif 1 <= diff <= 2:  use M = 4
elif 3 <= diff <= 6:  use M = 8
elif 7 <= diff <= 14:  use M = 16
base = floor((y-1)/M) * M
threshold = base + M - 1
if x < threshold:  choose (base + M/2 - 1, base + M/2)
else:  choose (base + M - 1, base + M)
```

The important point is not the specific integers themselves, but the design intent: overlap is resolved in a deterministic way that preserves correctness while nudging the final boundaries onto simple modulo classes. That is why the optimized selector plots look cleaner than the basic plots.